

Summary Data on Acid Cropland and Lime Requirements in Zambia

GAIA

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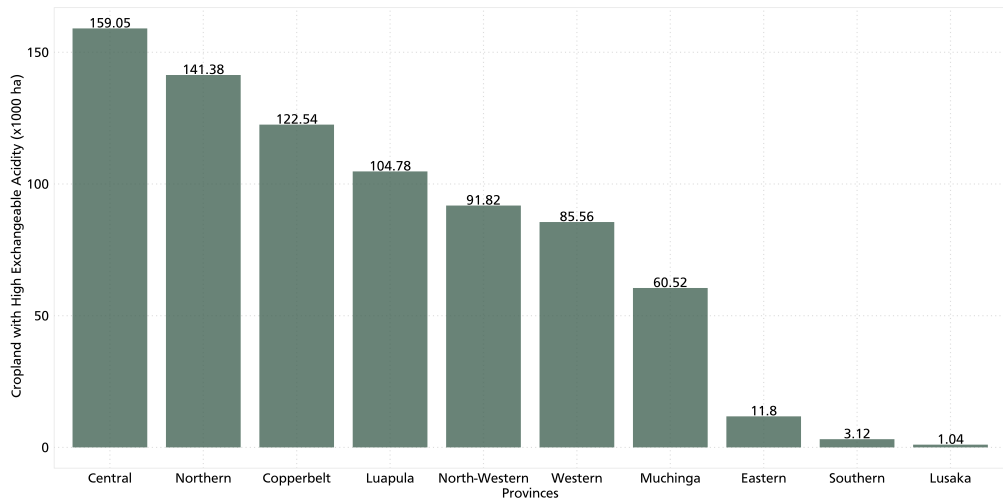
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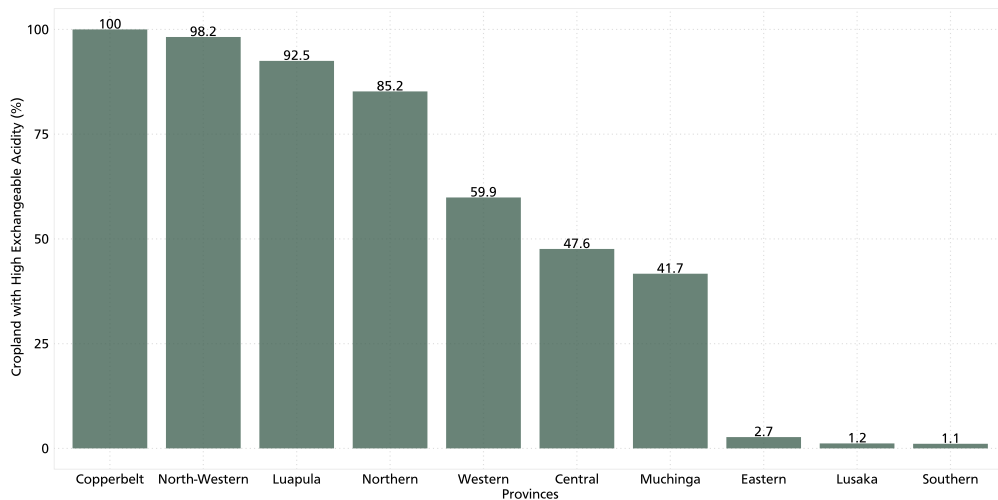
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Zambia**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production and production value from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

Top 10 Provinces in Zambia with the highest acidic cropland



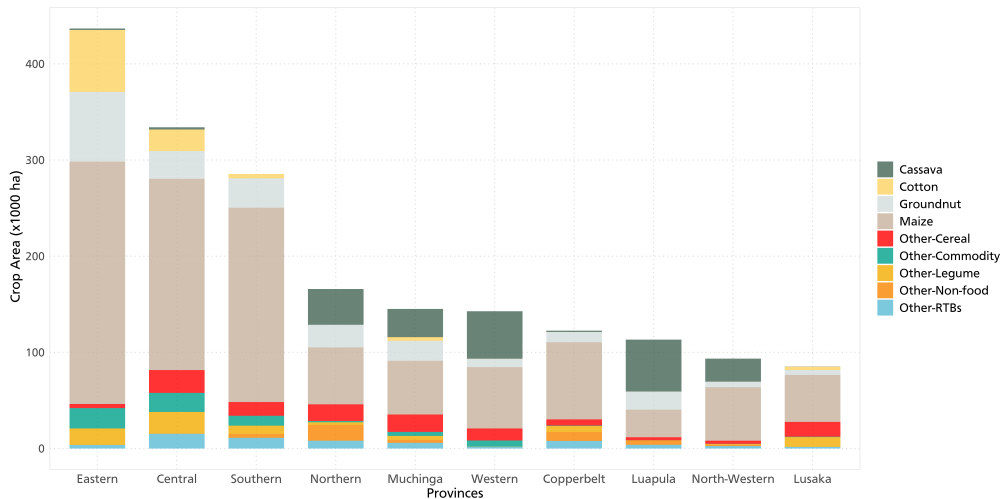
Top 10 Provinces in Zambia with the highest acidic cropland (Percentage of total cropland)



└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland



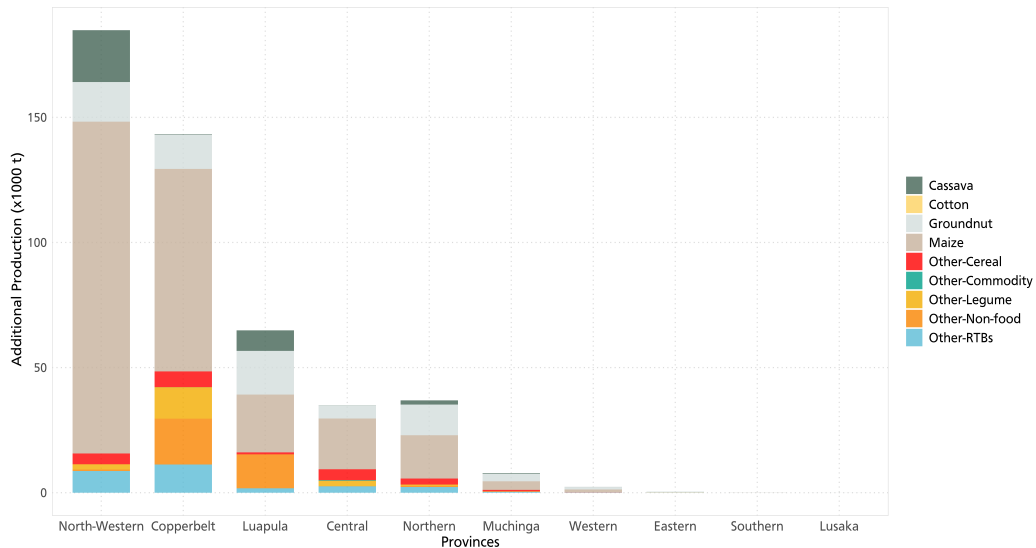
└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland (table)

Provinces	Cassava	Cotton	Groundnut	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-Non-food	Other-RTBs
Central	2.34	22.37	28.90	198.87	23.77	19.84	22.56	0.10	15.34
Copperbelt	1.34	0.08	10.61	80.24	6.38	0.50	6.30	9.21	7.88
Eastern	1.53	64.49	72.46	252.01	4.28	21.19	17.10	0.03	3.71
Luapula	53.98	0.03	18.78	28.68	3.16	0.06	0.38	4.31	3.85
Lusaka	0.23	3.32	5.11	48.92	15.32	0.42	10.20	-	1.72
Muchinga	29.36	3.75	20.84	55.85	18.12	4.15	4.12	3.13	5.86
North-Western	23.98	0.02	5.71	55.65	3.24	0.14	0.56	1.57	2.61
Northern	37.24	0.01	23.50	59.24	17.29	1.32	2.67	16.52	8.12
Southern	0.05	4.47	30.62	202.04	14.35	10.26	8.69	3.97	11.09
Western	49.36	0.09	8.66	63.77	12.52	6.35	0.18	-	1.81

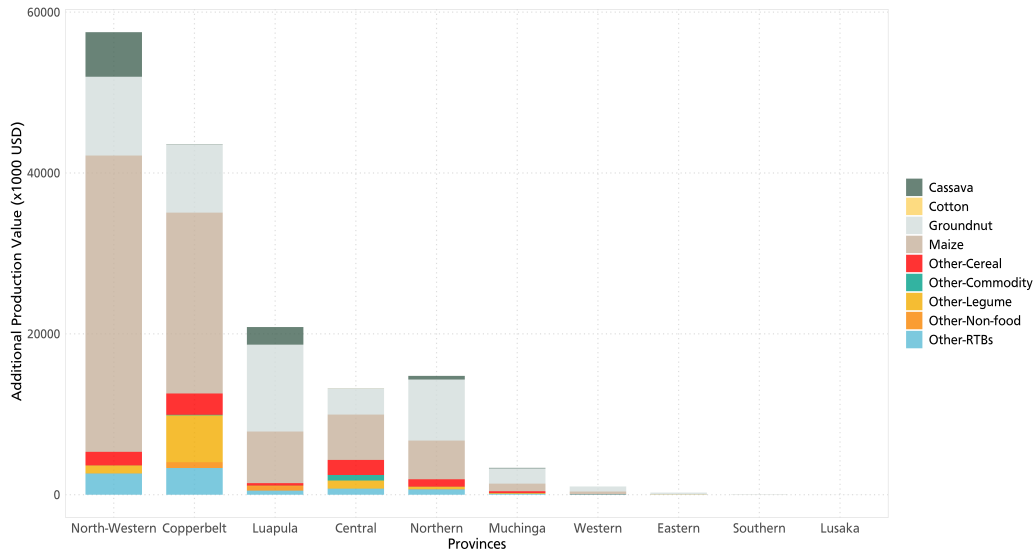
Potential for additional production (tone) in top 10 acidic Provinces in Zambia



Potential for additional production (1000 tone) in top 10 acidic Provinces in Zambia(Table)

Provinces	Cassava	Cotton	Groundnut	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-Non-food	Other-RTBs
Central	0.05	0.02	5.20	20.30	4.35	0.23	2.17	0.05	2.57
Copperbelt	0.19	0.01	13.59	80.94	6.26	0.02	12.60	18.30	11.28
Eastern	-	-	0.34	0.08	-	-	0.03	-	0.02
Luapula	8.19	-	17.45	23.10	0.82	-	0.12	13.43	1.75
Lusaka	-	-	0.01	-	-	-	-	-	-
Muchinga	0.25	-	3.05	3.42	0.60	-	0.18	-	0.35
North-Western	20.74	-	15.83	132.59	4.33	-	2.10	0.59	8.67
Northern	1.68	-	12.26	17.32	2.41	-	0.53	0.44	2.27
Southern	-	-	0.09	0.01	-	-	-	-	-
Western	0.03	-	1.01	1.13	0.05	0.01	-	-	0.08

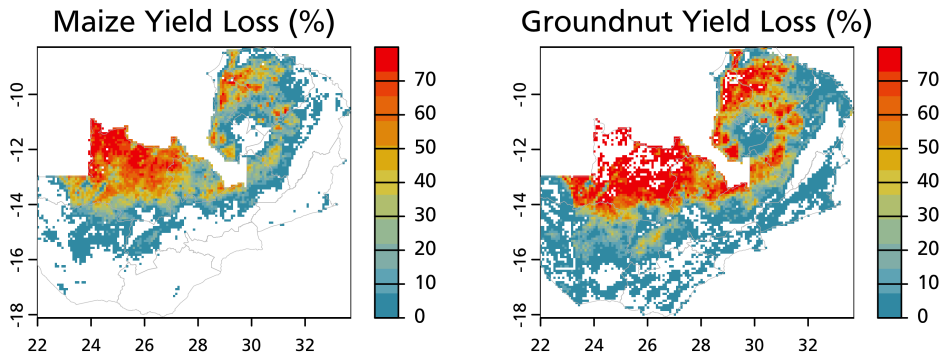
Potential for additional production value (US\$) in top 10 acidic Provinces in Zambia



Potential for additional production value (1000 US\$) in top 10 acidic Provinces in Zambia(Table)

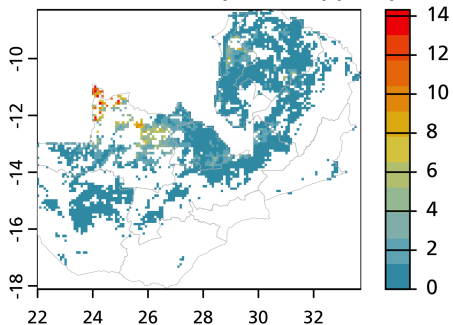
Provinces	Cassava	Cotton	Groundnut	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-Non-food	Other-RTBs
Central	14.5	9.2	3221.2	5638.3	1877.2	690.4	1003.6	2.0	754.6
Copperbelt	51.3	3.2	8419.3	22486.3	2650.9	63.5	5829.8	719.2	3318.7
Eastern	-	0.2	210.6	21.7	0.7	0.6	16.9	-	7.6
Luapula	2187.1	0.2	10807.6	6417.6	303.0	9.1	64.2	549.1	510.6
Lusaka	-	-	6.8	-	-	-	-	-	-
Muchinga	66.3	0.1	1892.4	949.3	229.0	6.1	91.3	-	103.3
North-Western	5537.4	0.1	9807.2	36832.5	1690.0	7.5	971.6	23.1	2639.4
Northern	449.4	-	7595.1	4810.2	912.7	9.0	288.6	37.4	666.7
Southern	-	-	54.7	1.9	-	-	-	-	-
Western	8.2	0.1	624.1	312.9	19.6	36.2	1.7	-	22.8

Yield loss due to soil acidity for Maize and Groundnut in Zambia

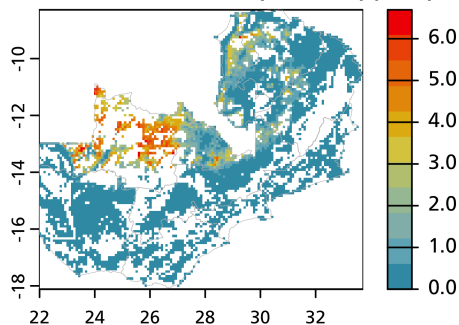


Yield response to lime application in Zambia

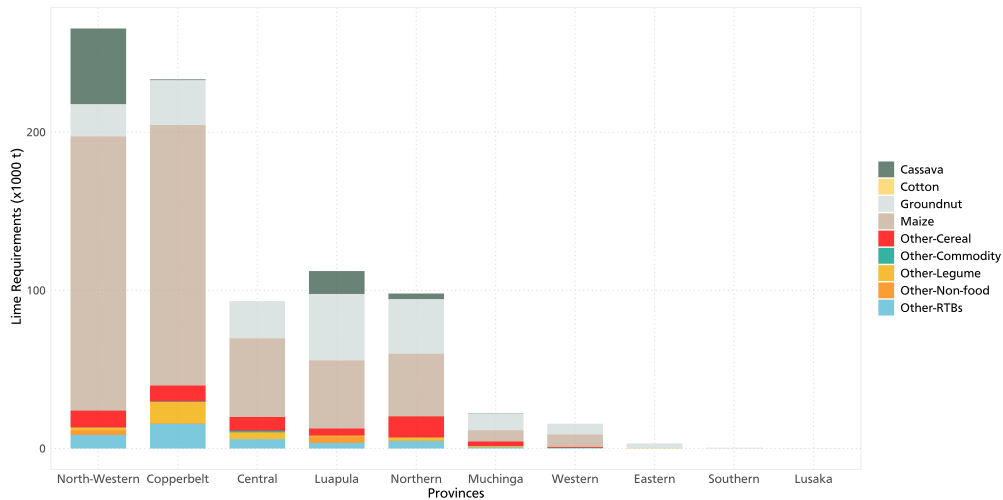
Maize Yield Response (t/ha)



Groundnut Yield Response (t/ha)



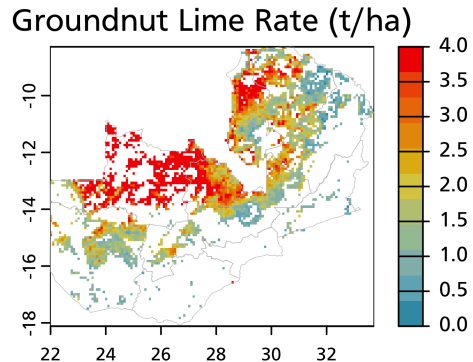
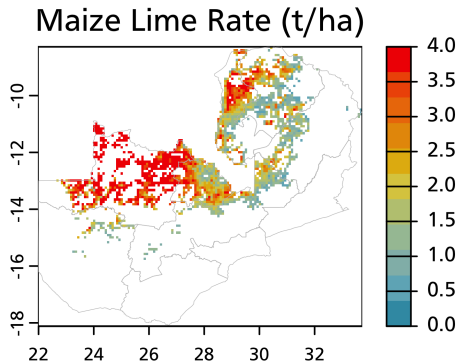
Lime requirements (t) for acid soil remediation in Zambia



Lime requirements (1000 t) for acid soil remediation in Zambia (table)

Provinces	Cassava	Cotton	Groundnut	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-RTBs	Other-Non-food
Central	0.06	0.04	23.09	49.72	8.76	0.95	4.27	5.96	-
Copperbelt	0.47	0.04	28.41	164.68	10.04	0.31	13.57	15.65	0.26
Eastern	-	-	3.11	-	-	-	0.13	-	-
Luapula	14.48	-	41.98	43.08	4.44	0.02	0.50	3.57	4.12
Lusaka	-	-	0.18	-	-	-	-	-	-
Muchinga	0.15	-	10.78	7.07	2.90	-	0.69	0.86	-
North-Western	47.79	-	20.48	173.35	10.61	0.06	1.86	8.63	2.81
Northern	3.54	-	34.55	39.61	13.30	0.04	2.00	4.75	0.24
Southern	-	-	0.53	-	-	-	-	-	-
Western	-	-	6.81	7.86	0.52	0.14	0.05	0.29	-

Lime rate for Maize and Groundnut in Zambia



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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